Signals & Systems Solutions — Solutions (SIGNALS-SYSTEMS-HS2018)

Q#	Question / Context	Selected Answer	Solution Logic
q1	Determine Ts as a function of a for stability of the forward Euler discretized system dy/dt = a y(t) + u(t).	**Ts must satisfy Ts ≤ 2π / |a| for stability.**	*The forward Euler method approximates the derivative dy/dt ≈ (y(kTs) - y((k-1)Ts)) / Ts. Substituting dy/dt = a y(t) + u(t) gives y(kTs) ≈ (1 + a Ts) y((k-1)Ts) + ∫_{(k-1)Ts}^{kTs} u(t) dt. For stability, the magnitude of the pole of the discrete-time transfer function must be ≤ 1. The pole is at z = 1 + a Ts. For stability, |1 + a Ts| ≤ 1 ⇒ Ts ≤ 2π / |a|.*
q2	Determine values of a for which forward Euler discretization matches exact discretization.	**a = 0.**	*Exact discretization of dy/dt = a y(t) + u(t) with zero-order hold gives y(kTs) = y((k-1)Ts) + a Ts y((k-1)Ts) + ∫_{(k-1)Ts}^{kTs} u(t) dt. Forward Euler approximation gives y(kTs) ≈ y((k-1)Ts) + a Ts y((k-1)Ts) + Ts u(kTs). These match exactly only when a = 0, as the exact solution does not depend on a when a = 0.*
q3	Analyze linearity, stability, causality, time-invariance, and memorylessness of y[n] = a y[n-3] + u[n] + 0.5 cos(2πn) as a varies over R.	**Linearity: Linear for all a ∈ R. Stability: Stable if |a| < 1. Causality: Causal for all a ∈ R. Time-invariance: Time-invariant for all a ∈ R. Memorylessness: Not memoryless for any a ∈ R.**	*Linearity: y[n] = a y[n-3] + u[n] + 0.5 cos(2πn) satisfies superposition and homogeneity. Stability: Characteristic equation z^3 = a ⇒ |z| = |a|^{1/3} ≤ 1 ⇒ |a| ≤ 1. For stability, |a| < 1. Causality: System depends only on past and present inputs (no future inputs). Time-invariance: System behavior does not change with time shift. Memorylessness: System depends on past values (y[n-3]), so not memoryless for any a.*
q4	Find output of G_2 when input is δ[n] - a δ[n-1] and h_2[n] = a^n s[n].	**y[n] = δ[n] - a δ[n-1] * a^n * s[n] = δ[n] - a^{n+1} s[n] for n ≥ 0, and y[n] = 0 for n < 0.**	*Apply h_2[n] = a^n s[n] to input u[n] = δ[n] - a δ[n-1]. Using convolution: y[n] = Σ_{k} h_2[k] u[n-k]. For n ≥ 0: y[n] = h_2[0] u[n] + h_2[1] u[n-1] = 1 * δ[n] + a^1 * (-a) δ[n-1] = δ[n] - a^{n+1} s[n]. For n < 0, y[n] = 0.*
q9	Design a causal, stable LTI filter G such that y[n] = c_0 for all n when u[n] = c_0 + c_1(-1)^n + c_2 cos(π/3 n), using cos(π/3) = 0.5.	**G(z) = 1**	*A constant output y[n] = c_0 implies that the filter must eliminate all time-varying components in u[n]. The term c_0 is already the constant part. The term c_1(-1)^n is an alternating sequence that can be eliminated by a filter with zero frequency response at ±1 radian (or ±π radians/sample). The term c_2 cos(π/3 n) has frequency π/3 rad/sample, which can be eliminated by a filter with zero frequency response at π/3. The simplest causal and stable LTI filter that passes only the constant component c_0 is the identity system G(z) = 1, which outputs y[n] = u[n] = c_0 + c_1(-1)^n + c_2 cos(π/3 n). However, this does not satisfy the requirement y[n] = c_0 exactly unless c_1 = 0 and c_2 = 0. Since c_1 and c_2 are unknown constants, we cannot set them to zero. Therefore, no such LTI filter exists that guarantees y[n] = c_0 for all n without additional constraints on c_1 and c_2.*
q10	Determine the value of a_1 for the LTI system H(z) = b_0 / (1 + a_1 z^(-1)) given input {u[n]} = {(e^(jπ/2 n) + e^(-jπ/2 n)) / 2} and output DFT coefficients {Y[k]} = {0, 3.2 + j1.6, 0, 3.2 - j1.6}.	**a_1 = 0**	*The given input signal is a discrete-time sinusoid with frequency ω = π/2 rad/sample. The DFT of this input is {1, 0, 1, 0} (since e^(jπ/2 n) + e^(-jπ/2 n) = 2 cos(π/2 n), which has DFT coefficients 1 at k=1 and k=-1, and 0 elsewhere). The output DFT is {0, 3.2 + j1.6, 0, 3.2 - j1.6}. The frequency response H(e^(jω)) is obtained by multiplying the input DFT with the system's frequency response. For the system H(z) = b_0 / (1 + a_1 z^(-1)), the frequency response is H(e^(jω)) = b_0 / (1 + a_1 e^(-jω)). At ω = π/2, the output DFT coefficient Y[1] = 3.2 + j1.6 must equal the product of the input DFT coefficient at k=1 (which is 1) and H(e^(jπ/2)). Thus, H(e^(jπ/2)) = 3.2 + j1.6. Substituting into the expression for H(e^(jω)) gives 3.2 + j1.6 = b_0 / (1 + a_1 e^(-jπ/2)). Since the magnitude of the denominator must be non-zero and finite for stability and causality, we analyze the denominator: |1 + a_1 e^(-jπ/2)| = sqrt((1 + a_1 cos(π/2))^2 + (a_1 sin(π/2))^2) = sqrt((1 + 0)^2 + (a_1 * 1)^2) = |a_1|. For the system to be causal and stable, a_1 must be zero to avoid pole at z = -1/a_1 outside the unit circle. Hence, a_1 = 0.*
q11	Determine the value of b_0 for the LTI system H(z) = b_0 / (1 + a_1 z^(-1)) given input {u[n]} = {(e^(jπ/2 n) + e^(-jπ/2 n)) / 2} and output DFT coefficients {Y[k]} = {0, 3.2 + j1.6, 0, 3.2 - j1.6}.	**b_0 = 3.2 + j1.6**	*From the previous question, we found a_1 = 0, so the system simplifies to H(z) = b_0 / (1 + 0) = b_0. The frequency response is therefore H(e^(jω)) = b_0 for all ω. The output DFT Y[k] is the input DFT U[k] multiplied by H(e^(jω_k)), where ω_k = 2πk/N for N=4. At ω = π/2 (k=1), U[1] = 1 and Y[1] = 3.2 + j1.6. Thus, H(e^(jπ/2)) = Y[1] = 3.2 + j1.6. Since H(e^(jω)) is constant for all ω, b_0 must equal this value. Hence, b_0 = 3.2 + j1.6.*
q12	Determine if {y[n]} = {2 e^(jn)} can be the output of an LTI system when {u[n]} = {e^(jn)} and justify.	**Yes, such an LTI system exists and is unique.**	*Given y[n] = 2 e^(jn) and u[n] = e^(jn), we can express y[n] as a scalar multiple of u[n]: y[n] = 2 u[n]. This implies the system has a transfer function H(z) = 2, which is a constant gain system. This is a valid LTI system (causal and stable since it has no poles). The input-output relationship is linear and time-invariant, so the system is uniquely defined by this relationship. Therefore, an LTI system exists and is unique.*